

Vulnerability Scanner (Mini Project)

Objective: Build a simple Python script to detect common vulnerabilities in a web application or network.

Key Features

- Scan for open ports using Python socket module.
- Identify weak configurations (e.g., default ports open).
- Display software/service banners when available to help identify outdated versions.
- Generate a simple vulnerability report in a text file.

Sample Python Program

```
import socket
target="127.0.0.1"
ports=[21,22,23,80,443,3306]
report=[]
for p in ports:
    s=socket.socket()
    s.settimeout(1)
    if s.connect_ex((target,p))==0:
        report.append(f"Port {p} is OPEN")
    s.close()
with open("vulnerability_report.txt","w") as f:
    f.write("\n".join(report))
print("Scan Complete")
```

Expected Outcome

The program scans common ports, identifies open ports that may indicate weak configurations, and generates a simple vulnerability report. This project provides basic exposure to penetration testing and vulnerability assessment concepts.